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Intrasubject Comparison of Four Response Modes for "Subjective Probability" Assessment[1]

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Despite much research on subjective probability or personal uncertainty (PU) little is known about how well various measures of PU agree on a within subject basis. Thirty Subjects made odds (O), probability (P), bets (B) and null datum (ND) responses to indicate their opinion about which of two normal distributions was being sampled. *O* and *P* responses were made verbally. For the B responses subjects played an active part in a betting procedure which allowed PU to be inferred. In the ND condition subjects responded by offsetting the probabilistic effect of a sample of two data with a third datum. PU was inferred from the offsetting datum. On both group and individual analyses the *O*, *P*, and *B* modes agreed well with one another and the ND mode differed significantly only from the *P* mode. Subjects may have used strategies to make their estimates but if so the strategies were in remarkable agreement.

Within the last decade a substantial amount of research has been carried out on the topic of "subjective probability" in dynamic situations. One reason for the interest shown in probability revision is the existence of an optimal model, Bayes' theorem, against which human performance can be compared. The typical, but not universal, result of such a comparison is to show that humans revise their "subjective probabilities" more conservatively, i.e., more slowly, than the optimal model. Before the results of such comparisons can be completely accepted, however, a more basic issue needs to be examined. What exactly does the subject's response represent? This same issue, of course, arises in other areas of psychology, e.g., magnitude estimation (Zinnes, 1969), but the focus of this paper is on "subjective probability." As Rapoport and Wallsten (1972) point out some researchers believe that there is "an identity relation between the response and its underlying psychological dimension" (p. 154) and others do not. (To help keep this important point in mind, and to avoid a bias towards any one response mode, this

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paper will label the underlying psychological dimension "personal uncertainty" or PU rather than subjective probability.) If an identity relation does not exist between PU and the response modes used then such findings as conservatism may turn out to be peculiar to the response mode chosen by the experimenter. Unless one is willing to simply assume that PU is identical to a particular response mode, e.g., probability, some way must be found to establish an identity empirically. Since we have no way of physically tapping into the human brain to measure PU we are forced, here as elsewhere, to rely on more indirect means of justifying our measures. One method is to seek agreement between the measures. If a large number of response modes which should be sensitive to PU yield comparable results then the measures can be assumed to relate meaningfully to PU. If agreement is lacking then some means of distinguishing which response modes relate to PU and which do not must be found.

A number of studies have been carried out examining the correspondence among various measures of PU. In accord with a distinction made by Beach and Wise (1969b) these studies can be broken down into those where PU is based on relative frequencies and those where the PU is about the likeliness of some hypothesis. Since the PU studied in the present experiment is of the latter sort only previous experiments of that kind will be reviewed here. Four studies are relevant. In one experiment Phillips and Edwards (1966) had subjects revise their PU about two binomial hypotheses on the basis of data randomly sampled from one of the binomial hypotheses (populations). Four different groups used four different response modes to indicate their PU revisions. The responses were given on scales of probability, log probability, odds and log odds. Differences were found among the response modes but since no indication of the data's variability is given one cannot tell if the differences are significant or if the response modes are in fact in essential agreement. In any case since different groups were used for each response mode nothing can be said about agreement between modes for individual subjects.

In another study which utilized group data Beach and Wise (1969a) found close similarity between probability estimates and confidence ratings. For one experiment the hypotheses were binomial and for the other multinomial. In both experiments two groups of subjects observed data sampled from one population (hypothesis) and said which population the sample was most likely to have come from. One group then gave a probability estimate for that population and the other group said how confident they were that their guess about the population was correct. Confidence was rated on a scale running from 50 (uncertain) to 100 (completely sure). Although a high degree of consistency was

found between these two response modes, from the present point of view there are two problems. First, the comparison again is between groups of subjects so no information is available on how well confidence ratings and probability estimates compare for individuals. Secondly, as the authors themselves point out, there is no set of numbers naturally associated with 'confidence.' A different scale might yield different results. The most that one can say is that when confidence is expressed on the scale Beach and Wise used, group measures of PU based on probability estimates agree with group measures of PU based on confidence ratings.

The other two studies of concern compared PU as measured by betting behavior to probability estimates. Such comparisons involve an additional complexity in that measures of PU can be obtained from betting behavior only by assuming the truth of some model (e.g., expected value) and then inferring PU on the basis of the model's predictions. Sanders (1968) had subjects indicate their PU about binomial hypotheses by making odds estimates and by choosing one bet from a list of bets about the hypothesis. The bets were generated by a quadratic payoff scheme which, under the assumption that subjects would maximize their subjectively expected values, allowed Sanders to infer a probability from the bet chosen. He then compared the estimated odds to the odds equivalents of the inferred probabilities and found them to be in good accordance. It is not possible, however, to say whether this agreement really represents individual subjects since a significant interaction was found between subjects and response mode (odds vs bets). In a somewhat similar study Beach and Wise (1969b) had subjects observe sequences of data bearing on the likeliness of two multinomial hypotheses. The subjects then estimated the probabilities of the hypotheses and bid for bets, offered by the experimenter, involving the hypotheses. Probabilities can be inferred from the bids by assuming that subjects will bid so as to maximize their subjective expected value. This assumption was weakened somewhat by the fact that the bets were hypothetical and no money changed hands. Beach and Wise also found good agreement between their two response modes. However, since no analyses were performed on individuals and since the group data had a high variance, no firm conclusion can be drawn as to how well the response modes agree for individuals.

It should be clear from the above that there is ample room for further work on the topic of response modes. The present study was undertaken to satisfy several goals. The first goal was to look at the agreement between a larger and more varied set of response modes for PU. The second was to try to overcome some of the artificiality of choosing a bet or bidding for a bet. The subjects in this experiment were more actively involved in determining the outcome of the gambles and real money

payoffs, albeit small ones, were used. The third goal was simply to look more closely at the issue of whether or not the consistency found between response modes represents just the group or the individual as well.

To satisfy these goals an experiment on PU revision using four different response modes was conducted. Two of the modes used the familiar probability and odds responses. Although these are the two main response modes used in the literature, no comparison has been made of their agreement on a within subject basis. The third mode used was a form of the bidding for a gamble technique which allowed the subject to interact with a computer to make his bid and play the gambles. The fourth mode was an attempt to measure PU without having the subject either gamble or make numerical estimates of PU. In this mode the subject acted as a sort of nulling instrument and gave his responses in stimulus rather than uncertainty terms.

METHOD

Subjects. Thirty male Rice University graduate and undergraduate students were individually run as subjects. They were paid $1.50/hr for most of the experiment; pay in the betting part of the experiment depended strictly on performance.

Apparatus and Stimuli. The experiment used two normal distributions as data generators. The distributions were the heights of United States men and women (see DuCharme, 1970 for details). The heights were displayed on a vertical surface marked off in feet and inches in units of ¼ in. Sliding pointers indicated the sampled heights. The random number sequence for the bet response mode was generated by a Digital Electronic Corporation PDP-8 computer coupled through a Memory Raster Display and presented on a CRT. The subjects used a keyboard linked to the computer to record their responses and to start and stop the random number sequence.

Response Modes and Instructions. Four response modes were used, odds (O), probability (P), bets (B), and null datum (ND). The O and P response modes have been extensively used and need little discussion. The subjects simply wrote down which population they thought the data sample had come from, followed by the probability or odds for that population.

Although basically simple the B mode task seems complicated at first exposure. Accordingly subjects were given two practice trials and extensive instructions which were elaborated and repeated as often as necessary. The reader is referred to Figure 1 as a visual aid for the following discussion of the B mode.

The subjects first indicated which population they thought a sample

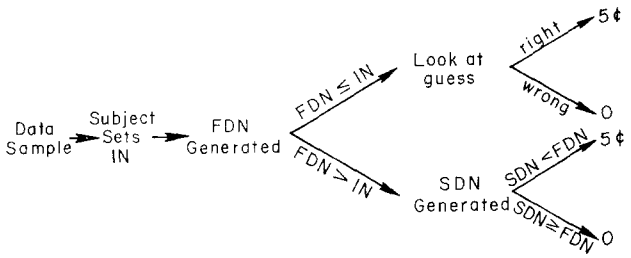

FIG. 1. A diagram of the sequence of events and outcomes in the *B* mode. See text for explanation of abbreviations.

of one or two heights had come from. They then named a bet such that they would be indifferent between playing the bet and being paid off on the basis of whether or not their guess about the population was right. The gambling device for the bets was the computer. It displayed on the CRT a sequence of random numbers between 0 and 999,999 at the rate of 10/sec (too fast to be individually apprehended). The subject could stop and start the random numbers by depressing a key on the computer-linked keyboard. The subject named a bet by choosing an indifference number (IN) between 0 and 999,999 and then pressed the key to "freeze" the first display number (FDN). If the FDN was equal to or smaller than the (IN) then no bet was played and the subject was paid 5¢ if his guess about the population was correct and nothing otherwise (upper branch of Figure 1). If the FDN was larger than the IN then a gamble was set in which numbers equal to or less than the FDN were winning numbers and those greater than FDN were losers. The gamble was played by having the subject start and then stop the random number sequence again to freeze a second display number (SDN). If SDN $\geq$ FDN the subject won nothing; if SDN $<$ FDN the subject won 5¢ (lower branch of Figure 1).

A concise statement of what a subject's strategy should be in this task is this—the surer he is that his guess is right the higher his IN should be. The reason for this is that he doesn't want to gamble with a smaller chance of success than his guess will give him. It can be shown that in order for the subject to maximize subjective expected value in this task he must set his IN so that IN/1,000,000 = PU. (The instructions included a verbal, intuitive proof of this point.) Under the maximizing assumption then the IN's can be transformed into a measure of PU. In summary the *B* mode offered subjects a means of expressing their uncertainty by choosing bets in a situation where they controlled the random gaming device.

In the ND mode the subjects saw only pairs of heights and responded

by naming a third height. Their instructions included three essential points (1) before seeing the two datum sample the male and female populations were equally likely, (2) after seeing the sample they would in most cases feel that one population was more likely than the other, and (3) that they should imagine a third height being drawn which would once again make the two populations equally likely. Their task was to name the third height which would nullify the diagnostic impact of the first two heights. In the ND response mode then no estimates of PU or bets were made, the subjects gave their responses in stimulus terms. For comparison with the other response modes the veridical odds equivalents of the nullifying heights were used in the data analysis.

Procedure. The first 24 subjects each went through the response mode conditions in a different order; the last six subjects went through in repetitious orders drawn at random, without replacement, from the 24 possible orders. The subjects were run individually in one session which lasted about two hours. Initially the subjects received some training on the attributes of the height populations. They first estimated the average height for the male population, saw a random sample of 25 male heights, made another estimate, saw another random sample of 25 heights and made their third and final estimate of the mean. This sequence was then repeated for the female population. By the third estimate the average subject was off by less than ⅛ in. on both means. A total of 30 different height samples were used in the main part of the experiment. Twenty of these samples consisted of a single height and 10 were pairs of heights. Half of the samples favored the male population and half the female. All 30 samples were used in the *O*, *P*, and *B* conditions but only the 10 pairs of heights were used in the ND condition. Single height samples were avoided in this condition because they might have been too suggestive of a simple response strategy which will be discussed later in the paper.

Prizes of $5 were offered for the best performance by a subject in the *O*, *P*, and ND conditions. "Best" was defined for the subjects as being closest to the optimal model. The prizes were offered to provide a monetary incentive in these conditions comparable to the one existing in the *B* condition. A questionnaire was given to each subject upon completion of the experiment asking him to describe *how* he arrived at his estimates in each response mode.

RESULTS

The first step in the data analysis was to transform all the responses to the common form of inferred log likelihood ratio (ILLR). The ILLR is a measure of the diagnostic impact assigned to a data sample by a

subject and was calculated for the various response modes in the following way:

$$\text{Odds } (O) \text{ ILLR} = \log \text{ estimated odds}$$

$$\text{Probability } (P) \text{ ILLR} = \log \frac{\text{estimated probability}}{1 - \text{estimated probability}}$$

$$\text{Bet } (B) \text{ ILLR} = \log \frac{\text{IN}/10^6}{1 - (\text{IN}/10^6)}$$

In the ND condition the ILLR was simply taken to be the log likelihood ratio which corresponded to the height stimulus given by the subject. The ILLR's were then compared to the veridical or Bayesian log likelihood ratios (BLLR) using a difference measure (D), where,

$$D = \text{ILLR} - \text{BLLR}.$$

Note that the ILLR and BLLR will be the same in the ND condition only when the subject has chosen a nullifying height which does in fact make the two populations equally likely.

Figure 2 shows the mean D values, calculated for each of the 30 subjects and then averaged over subjects, plotted for each response mode. The 99% confidence interval around each point is also displayed. The data analysis used 99% confidence intervals so that the probability of all four means being simultaneously covered by the intervals would be 0.96. Significance statements are thus at the 0.04 level; for details see Hays (1963, p. 313). In general the figure shows that there is good agreement among the response modes. The only confidence intervals which fail to overlap are those for conditions P and ND. The finding that the ND responses seem to be generally closer to optimal than the others may be due to an artifact which will be discussed later.

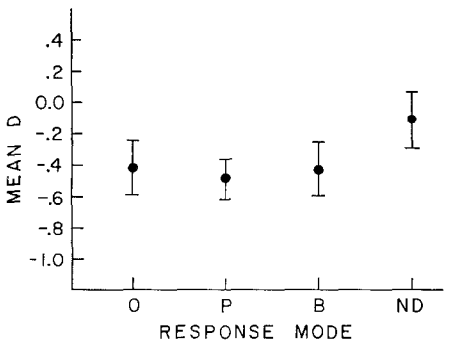

FIG. 2. Mean D for each response mode. Vertical lines represent 99% confidence intervals.

The same D analysis carried out for each subject indicates that the average responses are accurate reflections of what individuals were doing. Twenty-four of the 30 subjects have no significant differences between any of the response modes. Although the number of subjects showing significant differences is more than one would expect by chance (given the null hypothesis of no difference) the finding that 24 out of 30 do *not* show significant differences seems the more important fact. The most common difference which does occur, as might be expected from the mean results, is that the ND responses are significantly more optimal than the P responses.

A regression analysis of the average ILLR's and BLLR's yields the same general pattern of results as the D analysis. Pearson product moment correlations between ILLR and BLLR ranged from .95 to .99 with slopes of .54, .52, .57, and .78 for the O, P, B, and ND responses, respectively.

DISCUSSION

The experiment shows good accord between three of the response modes both on a group and an individual basis, but the ND responses are anomalous. Not only were the ND responses significantly different from the P responses, but there is also a clear trend for them to be closer to optimal than the O or B responses. What is different about the ND response mode? Although several explanations suggest themselves, one in particular merits discussion. It may be held that an artifact is present in the ND condition and that the amount of optimality displayed is due to this artifact. Slovic and Lichtenstein (1971) have argued that for a variety of reasons subjects often use simple strategies in inference tasks. The particular simple strategy which comes to mind in the ND task would be one where the subject first chooses a height which he thinks is just as likely to be from the male or female population and then makes his ND responses in terms of this "equally likely height" in the following way. If the height he has to null is three inches *below* the equally likely height his nulling response will be a height three inches *above* the equally likely height. If the subject chooses an equally likely height close to the actual one and applies this strategy consistently and accurately his responses will be very nearly optimal in this task. (Note that if the variances of the two populations are unequal this strategy will not work.) The strategy is most likely to occur to subjects when only a single height has to be nulled and it was for this reason that only samples of two heights were used in the ND condition. With two heights the strategy is not as obvious but could still occur. The subjects could have used the discrepancy between the equally likely height and the

mean of the sample heights or they could have used the *sum* of the discrepancies of the two sample heights from the equally likely height. Either strategy will work. One of the items on the postexperiment questionnaire sought evidence of simple strategies in all response conditions and examination of the questionnaires showed that eight subjects did in fact use an averaging strategy to determine their ND responses. However, a new data analysis excluding the eight subjects who averaged yielded the same conclusion as the full analysis; the ND responses are more optimal. Since it is possible that some of the subjects used the averaging strategy, but did not report doing so, this experiment can yield no clear-cut conclusion as to whether or not the ND response mode is truly different from the others. An experiment to test that notion would have to use data generators which do not lend themselves to simple strategies. One possibility would be to use complex data generators each of which contain both normally and binomially distributed data. The subject could be asked, as in the present experiment, to give a data sample which would offset the diagnostic impact of selected samples but he would be asked to offset a binomial sample with normal data or vice versa.

The fact that the *O* and *P* responses are in agreement means that one should be able to make ready comparisons between previous studies which have used these response modes. This ability becomes important when one realizes that nearly all previous studies of PU revision have used either odds or probabilities as responses.

The correspondence between PU inferred from the *B* responses and PU as measured by *O* and *P* responses is in accord with previous experiments where the subject's role in the gambles was more passive. As to what this agreement means it should be pointed out here that although the response in this type of task can be viewed as a choice among bets it can also be viewed as another number estimation task, albeit one of a different sort. While this observation does not decrease the importance of the agreement found between the response modes, it does leave open the question of how simple choices relate to PU. In this light a study by Wise (1970) is of interest because it showed that estimated probabilities and probabilities inferred from simple choices are consistent. In Wise's study subjects assigned probabilities to hypothetical probabilistic statements and also chose which statement in a pair was most probable for all possible pairs of the hypothetical statements. Probabilities were generated from the paired comparison data by means of Thurstones Law of Comparative Judgment. The two estimates of PU were found to be highly congruent.

In conclusion what can be said about response modes and PU? The

indication is that the response modes used in the present study, except perhaps for the ND mode, are all equally good measure of PU. However PU is encoded or realized internally these are all equally good methods of getting at it. If subjects are using simple strategies to generate their responses then these strategies are in remarkable agreement. It is possible that subjects do settle on a strategy initially and then use the strategy thereafter to relieve them of the task of making intuitive estimates repeatedly, but if so they must be checking the fit of the initial results of the strategy against some internal representation. Only in this way could such widely different response modes yield similar estimates of PU.

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